CA-006

Problem: In a digital world where documents can be easily modified there is need for method to certify when document was last changed. Traditional methods like notary stamps and mailing documents to self are not feasible.

Solution: Haber and Stornetta propose computational method for digital time stamp which ensures documents will not be forward-dated or backdated even with collusion with Time-Stamping service. This approach maintains privacy and does not require permanent record keeping

Key Concepts

Hash functions -> unique hash value, a fixed length fingerprint is time-stamped preserving privacy & storage needs

Digital Signatures -> Trusted time stamping service signs document hash with timestamp

Linking -> time-stamps are chained together so that each new timestamp references previous one. this will prevent fraud as fake timestamps would require breaking hash

Distributed trust -> Group of randomly chosen users sign the timestamp, making forgery impractical

It can be used in

IP protection of previous work

Business records to maintain integrity of logs

Digital media authenticity to image, videos etc

financial transactions (prevents tampering sequential records)